Huawei Access Node KR-ACC-15 Optical Transport Failure SOP

Node: KR-ACC-15 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_2659

Description

This Standard Operating Procedure (SOP) addresses a specific failure domain of Optical Transport Failure on the Huawei Access Node KR-ACC-15 at the KR Puram Access site. This failure domain is primarily attributed to the degradation of SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss. The Optical Transport Network (OTN) protocol, which operates at the data plane, facilitates the transmission of client data over the OTN network. However, various hardware components, including SFP/XFP transceivers, optical fibers, and transponders, can contribute to the failure of the OTN protocol. In the Huawei Access Node KR-ACC-15, the OTN protocol relies on the Optical Network Unit (ONU) to receive and transmit client data. The ONU is connected to the optical fiber, which carries the client data. If the SFP/XFP transceiver degrades, the Rx/Tx power loss increases, or a fiber cut occurs, it can lead to a failure of the OTN protocol, resulting in Alarm Code ALARM_2659. The OTN protocol is responsible for multiplexing and demultiplexing client data into a single wavelength, which is transmitted over the optical fiber. Therefore, any impairment in the OTN protocol can lead to a failure of the Access Node KR-ACC-15.

Pre-requisite Checks

- Pre-Condition: Verify the site location and node ID to ensure accuracy, Description: Confirm that the failure has occurred on the Huawei Access Node KR-ACC-15 at the KR Puram Access site

- Pre-Condition: Check the redundant link status, Description: Verify that the redundant link is operational to ensure continuous network availability

- Pre-Condition: Dispatch remote hands to KR Puram Access, Description: Ensure that remote hands are dispatched to the site for physical checks and repairs

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display otn transponder info, Expected Output: OTN Transponder Information, Description: This command displays the OTN transponder information, including the transponder ID, type, and status. It helps to identify if the transponder is operational or if there are any issues with the SFP/XFP transceiver.
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```
Command: display ont transceiver info, Expected Output: ONT Transceiver Information, Description: This command displays the ONT transceiver information, including the transceiver ID, type, and status. It helps to identify if the ONT transceiver is operational or if there are any issues with the optical fiber.

Command: display otn alarm info, Expected Output: OTN Alarm Information, Description: This command displays the OTN alarm information, including the alarm code, description, and clear status. It helps to identify the cause of the alarm and take corrective action.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceiver, Command: replace otn transponder 1/1/1-1/1/1 with transceiver 1/1/1-1/1/1, Description: Replace the SFP/XFP transceiver with a new one to resolve the transceiver degradation issue.

Step 2: Step: Repair or replace the optical fiber, Command: replace otn transceiver 1/1/1-1/1/1 with optical fiber 1/1/1-1/1/1, Description: Repair or replace the optical fiber to resolve the fiber cut issue.

Step 3: Step: Adjust the Rx/Tx power settings, Command: adjust otn transceiver 1/1/1-1/1/1 power settings, Description: Adjust the Rx/Tx power settings to resolve the power loss issue.

Post-Check Verification

```
Command: display otn transponder info, Expected Output: OTN Transponder Information, Description: Verify that the transponder is operational and there are no issues with the SFP/XFP transceiver.

Command: display ont transceiver info, Expected Output: ONT Transceiver Information, Description: Verify that the ONT transceiver is operational and there are no issues with the optical fiber.

Command: display otn alarm info, Expected Output: OTN Alarm Information, Description: Verify that the alarm has been cleared and there are no new alarms.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: restore otn transponder 1/1/1-1/1/1 with original transceiver, Description: Restore the original SFP/XFP transceiver if the replacement does not resolve the issue.

- Step: Restore the original optical fiber, Command: restore otn transceiver 1/1/1-1/1/1 with original optical fiber, Description: Restore the original optical fiber if the repair or replacement does not resolve the issue.

- Step: Restore the original Rx/Tx power settings, Command: restore otn transceiver 1/1/1-1/1/1 power settings, Description: Restore the original Rx/Tx power settings if the adjustment does not resolve the issue.